Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Project Report | Unified Mentor / Parcl Co. Limited

1. Executive Summary

This project explores buyer segmentation using client demographics, acquisition purpose, financing indicators, satisfaction scores, and linked sold-property activity. The supplied datasets contain 2,000 client records and 10,000 property records. K-Means was run with four clusters as directed by the project brief, with a sampled Hierarchical Clustering comparison. The K-Means sampled silhouette score was 0.125; the Hierarchical sample score was 0.098. These modest values suggest the segments overlap and should be treated as exploratory rather than definitive market categories.

2. Problem Statement and Objectives

The project brief states that Parcl lacks a data-driven understanding of buyer types, investment motivations across demographics, geographic differences in investment behaviour, and customer financing patterns. Objectives are to clean and combine the datasets, explore patterns, cluster buyers, compare clustering approaches, and present results in an interactive dashboard.

3. Data Description and Preparation

clients.csv contains client_id, client_type, first_name, last_name, date_of_birth, gender, country, region, acquisition_purpose, satisfaction_score, loan_applied, and referral_channel. properties.csv contains listing_id, tower_number, transaction_date, unit_category, unit_number, floor_area_sqft, sale_price, listing_status, and client_ref. Client data had no missing values in the supplied file. The property client_ref field had 2,695 missing values; these records are not attributed to a client during aggregation. Sale prices were converted from currency-formatted text to numeric values. Sold property records were aggregated by client_ref and left-joined to client profiles. Age is approximated as of 12 October 2026.

4. Methodology

Numeric features (age, satisfaction score, sold-property count, total purchase value, average purchase price, average area, and number of property types) were median-imputed and standardized. Categorical features (client type, gender, country, region, acquisition purpose, loan application, and referral channel) were imputed with the most frequent value and one-hot encoded. K-Means used k=4, random_state=42, and n_init=10. Hierarchical clustering with Ward linkage was evaluated on a reproducible sample of up to 500 records. Silhouette scores were estimated on samples to manage runtime.

5. Results

Cluster IDs are algorithm-generated labels and do not automatically correspond to the example buyer types listed in the project brief. The table below summarizes observed profiles; interpret them from measured values rather than assigning unsupported labels such as “Global Investor” or “Luxury Investor.”

Cluster	Buyers	Mean age	Satisfaction	Investment purpose	Loan applied	Avg. purchase value	Sold properties
C1	586	56.3	2.91	31.7%	38.6%	$311,342	2135
C2	49	68.1	3.49	34.7%	32.6%	$334,390	360
C3	708	54.3	3.07	30.5%	35.0%	$305,371	2487
C4	657	55.4	3.06	29.8%	37.4%	$424,879	2323

Key observations

• Cluster C2 is the smallest segment (49 clients) and has the highest mean age (68.1) and satisfaction (3.49). • Cluster C4 has the highest average purchase value among clients with linked sold-property records ($424,879). • Investment-purpose buyers make up approximately 29.8%–34.7% of each cluster, so acquisition purpose alone

does not sharply separate these clusters. • The K-Means silhouette score (0.125) is low, indicating substantial overlap; further feature selection and cluster-count tuning may improve interpretability.

6. Dashboard Design

The Streamlit dashboard provides a buyer-segmentation overview, investment-behaviour charts, geographic analysis, descriptive cluster profiles, client-level records, and CSV export. Sidebar controls filter by country, region, acquisition purpose, and client type. The dashboard is intended for exploratory analysis and does not make investment recommendations for individual buyers.

7. Recommendations

1. Treat the four clusters as exploratory segments and validate them with business stakeholders before using them in marketing campaigns. 2. Investigate the very small C2 cluster and check whether it reflects a meaningful buyer group or a feature-scaling / categorical-encoding effect. 3. Improve data linkage for property records with missing client_ref values before drawing client-level purchase conclusions. 4. Evaluate alternative feature sets, k values, and distance metrics; report silhouette scores alongside business interpretability. 5. Use country, region, acquisition purpose, and client type filters to support targeted analysis, while avoiding claims of causation.

8. Limitations

The dataset does not include an explicit income field, although the project brief’s suggested segment descriptions refer to income and “large investments.” Those labels cannot be verified from the available fields. Property linkage is incomplete for 2,695 rows. The silhouette scores are sampled and should not be compared as a formal model-selection study without consistent validation procedures. The report is based on the supplied datasets and project brief.

9. Conclusion

The workflow combines client profiles with sold-property activity and demonstrates an end-to-end approach to buyer segmentation, model comparison, and interactive market analysis. Results show differences in age, satisfaction, and average purchase value across the four clusters, but the low silhouette score indicates that the current segments overlap. Better client-property linkage, feature refinement, and business validation are recommended before operational use.

Reproducibility: run `python analysis.py` to regenerate cluster outputs, and `streamlit run app.py` to launch the dashboard.